Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **RetailNova**
b. Team leader name: **Suvetha S**
c. Problem Statement: **Retail & Commerce: Intelligent Customer and Business Experiences**

RetailNova

Every conversation is a demand signal.

Gemini agents on Google Cloud that turn what shoppers ask for into what stores stock.

Team: Suvetha S (lead) · Manoj T

Brief about the idea

The problem: stockouts hide demand

Retailers forecast from what they sold, not from what customers wanted. When an item is out of stock, its demand disappears from the data: zero sales → a smaller forecast → a smaller order → the gap repeats every season.

The shopper who asked for a maroon silk saree before Diwali and walked away is never counted.

$1.2T

lost to out-of-stocks globally each year

IHL Group, 2025

37%

of shoppers go to another store when an item is unavailable

IGD ShopperVista

₹5.4L cr

Navratri → Diwali 2025 trade; ~85% via physical retail

CAIT, Oct 2025

RetailNova closes the loop

Ask — Shoppers ask in their language by text, photo or voice note; staff log walk-in requests.

Serve — A Gemini concierge answers only from the real catalog and live stock in every store.

Sense — Each conversation becomes a structured demand signal: what, where, price, and why it was unmet.

Plan — BigQuery TimesFM forecasts plus signals → an agent proposes transfers and reorders, with evidence.

Act — The manager approves; inventory moves; the waiting shopper is notified.

Built for Indian festive retail: physical stores + online, 21 days before Diwali (8 Nov 2026). All demo data is synthetic; Nova Bazaar is a fictional retailer.

Opportunities

How is it different?

Shopping assistants stop at the conversation. Forecasting tools only see sales.

RetailNova connects them: demand that customers voice becomes a planning input the same day.

Google's Conversational Commerce agent serves the shopper side; RetailNova closes the loop to store operations.

How does it solve the problem?

Captures unmet demand when it is expressed: chat, photo search, store assist.

Turns it into structured signals, then a demand-sensed TimesFM forecast.

An agent proposes evidence-backed transfers and reorders; a human approves; the shopper is notified.

USP

Demand you can't see in sales, made visible and actionable.

Measured (synthetic backtest): forecast error 34.3% → 33.0%; lost revenue cut by 11.1%.

Grounded: 100.0% of recommended products verified against catalog and stock.

List of features offered by the solution

Multilingual concierge

Text, photo or voice note in Tamil, Hindi or English. Grounded in catalog + live stock; never invents products or prices.

Nova remembers you

Agent Platform Memory Bank recalls taste and budget across visits. One-tap “Forget me”.

Omnichannel availability

Stock across 6 stores + online. Reserve with a QR pickup pass or tap “Notify me”.

Every ask is a signal

Chats, voice notes, photos and walk-in requests logged by staff become structured demand signals.

Demand Radar

Sales-only vs demand-sensed TimesFM forecasts, assortment gaps and revenue protected.

Agentic action plan

Parallel steps incl. live Google Search market trends → transfers and reorders with evidence.

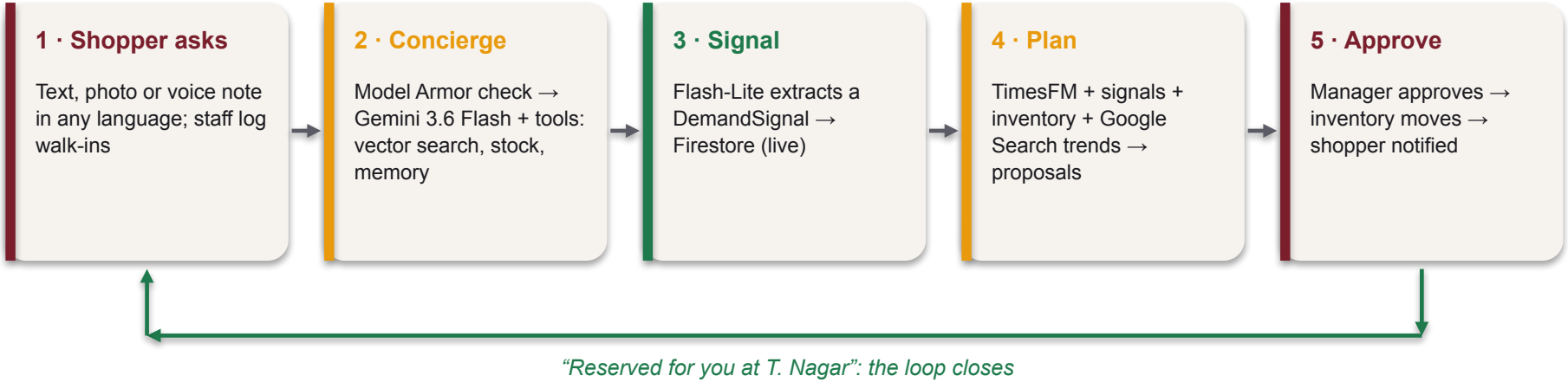
Approve → notify

One click moves stock and notifies waiting shoppers. Morning brief for the manager.

Guardrails

Model Armor screens every prompt; server-side grounding; rate limits; no personal data.

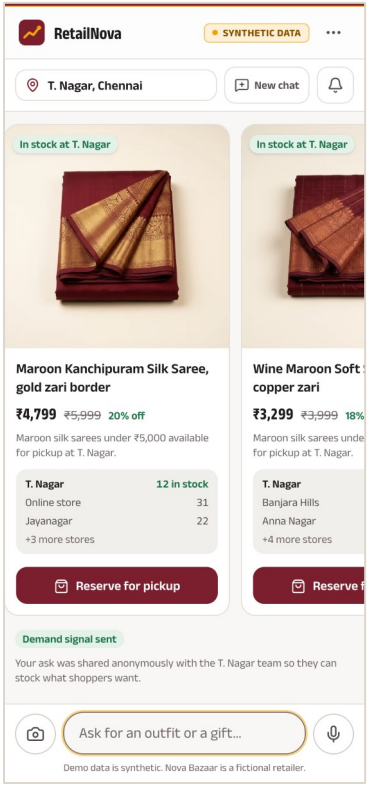
Process flow diagram

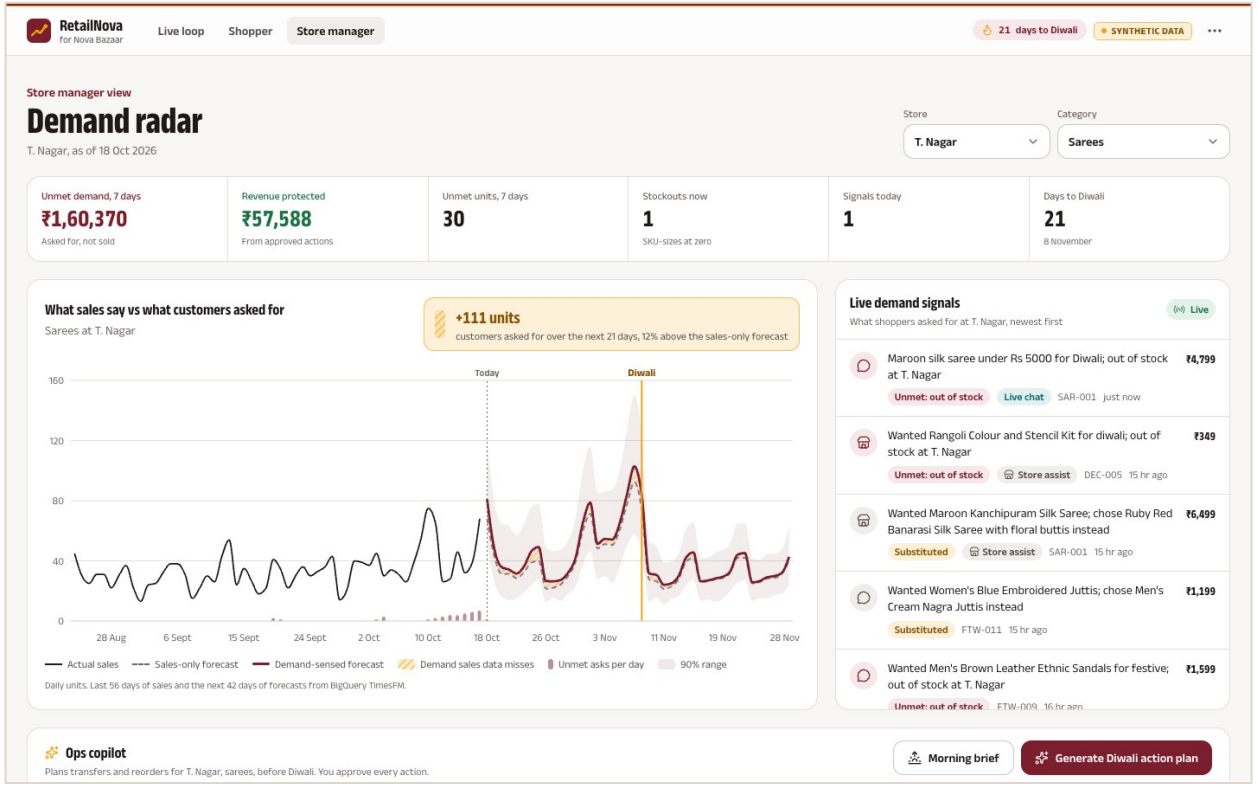


Use case (hero flow)

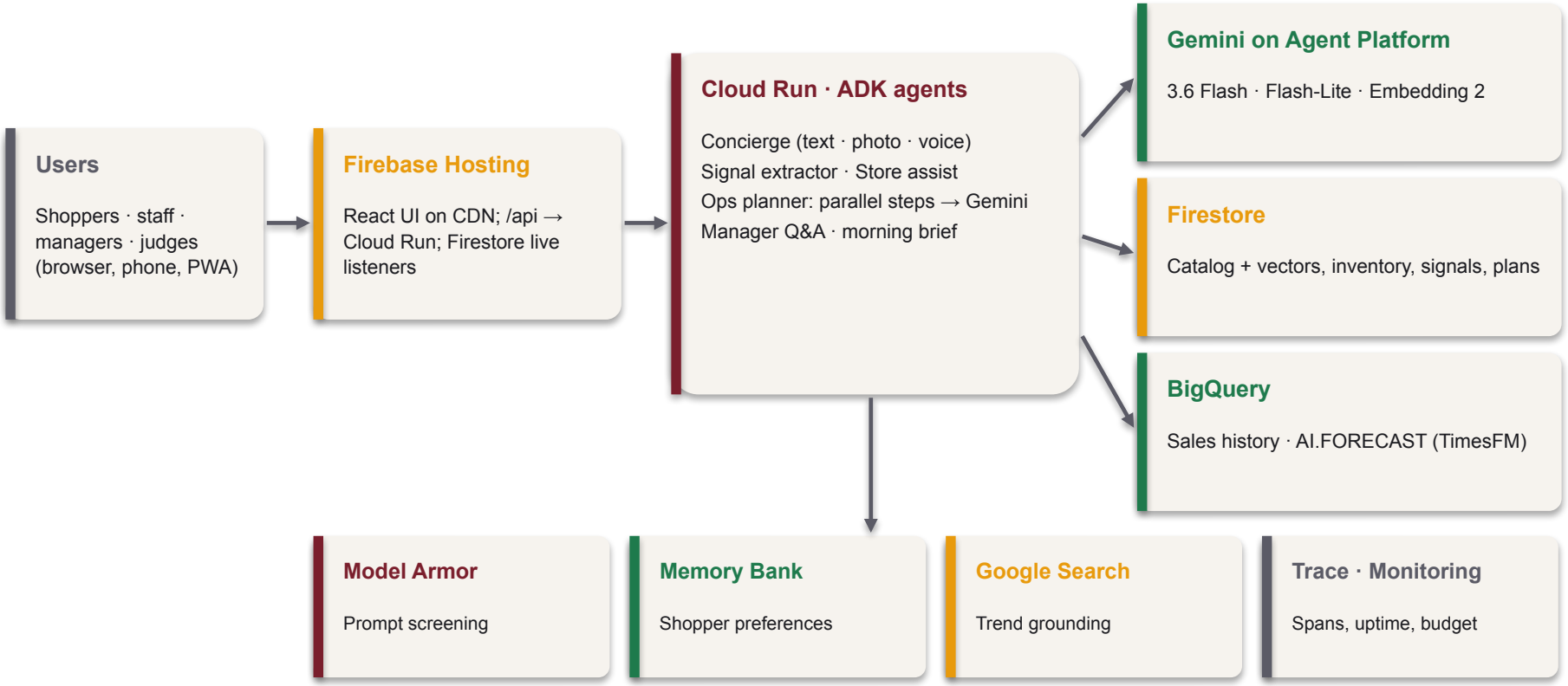
Priya asks for a maroon silk saree under ₹5,000 for Diwali pickup at T. Nagar. It is out of stock there but 14 sit in Coimbatore, so she taps "Notify me". Arjun's Radar shows the unmet-demand spike live; the plan proposes a transfer from Coimbatore plus a reorder; he approves; Priya's chat shows it is reserved for her.

Wireframes / mock diagrams of the solution





Architecture diagram of the proposed solution



Technologies used in the solution

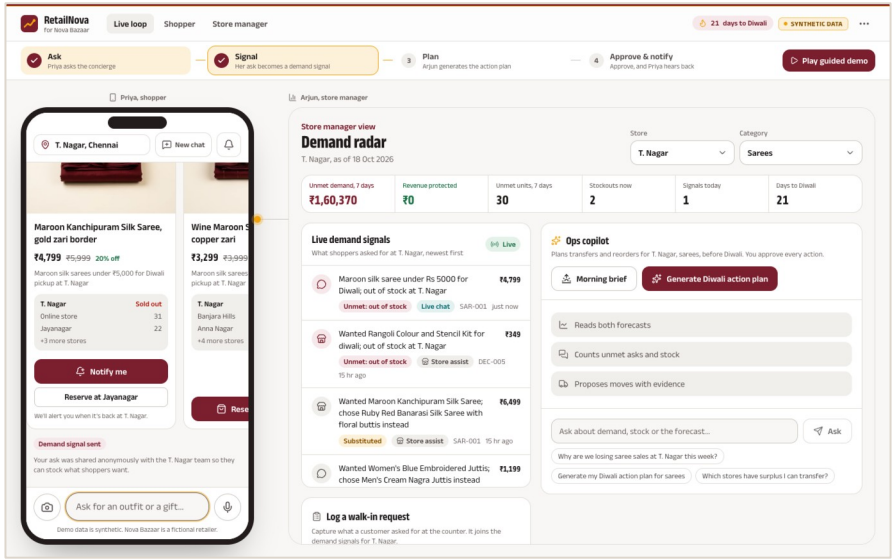
Google service	Used for
Gemini 3.6 Flash (Agent Platform)	Concierge reasoning + tool calling, voice-note and image understanding, multilingual chat, ops planning, morning brief
Gemini 3.5 Flash-Lite	Structured DemandSignal extraction (JSON schema); hedged fallback model
Google Search grounding	Live festive market trends with cited sources inside the action plan
Agent Platform Memory Bank	Remembers shopper preferences across visits (preferences only); “Forget me”
Model Armor	Screens every prompt for injection, jailbreak and harmful content before tools run
Gemini Embedding 2 · Gemini 3.1 Flash Image	768-dim vectors for semantic + photo search; generated the catalog photo set
Agent Development Kit (ADK)	Agents, function tools, sessions, streamed events
Cloud Run	Serverless agent API (FastAPI); autoscaling, keyless service identity
Firebase Hosting	Global CDN for the UI (installable PWA); /api rewrite to Cloud Run
Firestore	Catalog, inventory, signals, plans; vector search; real-time listeners in the UI
BigQuery + AI.FORECAST (TimesFM)	2-year sales history; forecasts with no model training; backtest analytics
Cloud Trace · Monitoring · Scheduler · Build	Spans per agent/tool call, uptime alerts, nightly demo reset, container builds

Estimated implementation cost

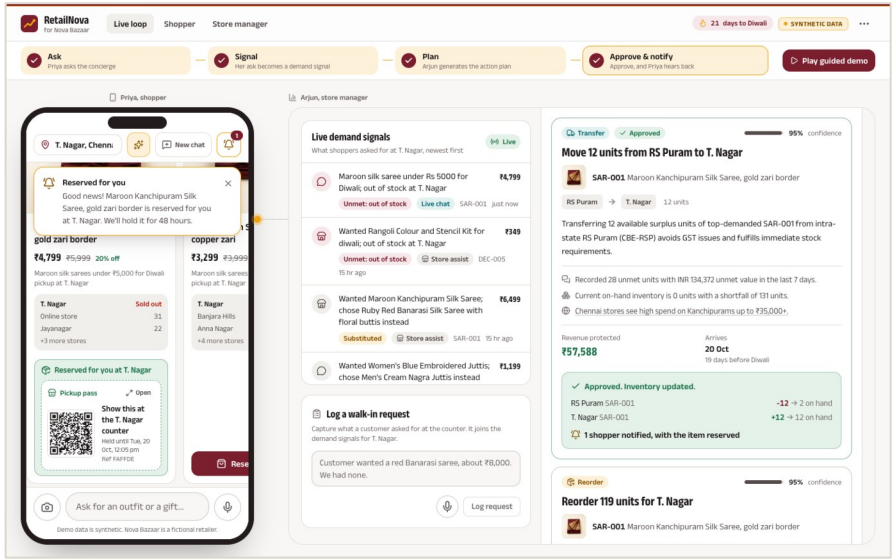
Item	Prototype to date (est.)	Pilot: 50 stores, 100k chats/month (est.)
Gemini (agents + extraction)	₹500	₹94,000 / month
Cloud Run + Firebase Hosting	₹100	₹4,000 / month
Firestore + BigQuery	₹50	₹3,000 / month
Total	₹650	₹101,000 / month

Measured cost per 1,000 shopper conversations: ₹940 (from real token usage on Gemini 3.6 Flash + Flash-Lite).
Serverless by design: Cloud Run scales to zero, Firestore and BigQuery bill per use, and no GPUs are needed (TimesFM runs inside BigQuery).
Payback logic: recovering even 1% of festive lost sales for a 50-store chain outweighs the platform cost (see the benchmark slide for simulated recovery).

Snapshots of the prototype



1 · Priya asks in Hinglish → grounded cards, sold out at T. Nagar → the demand signal lands live on Arjun's radar



2 · Arjun approves the plan: 12 units move from RS Puram → Priya's phone: “Reserved for you at T. Nagar”

Live at https://retailnova-ai.web.app (no login). Also: mobile shopper chat, photo search, Tamil, manager Q&A.

Prototype performance report / benchmarking

Backtest: Diwali 2025 window (synthetic, true demand known)

Metric	Sales-only forecast	Demand-sensed (+ signals)
Forecast error (WAPE)	34.3%	33.0%
Stockout days	159	141
Lost revenue (6 weeks)	₹125.9 lakh	₹111.9 lakh (−11.1%)

Across capture rates of 10-50%, demand sensing cuts lost revenue by 9.6-11.3% and WAPE by 1.1-1.4 points versus sales-only, with gains growing as more unmet demand is captured (simulation; capture rate assumed known).

Method: TimesFM (BigQuery AI.FORECAST), 42-day horizon, cutoff 6 weeks before Diwali; the same replenishment policy for both arms, scored against generator ground truth.

Concierge and system

Measure	Result
Grounded products (no invented SKUs)	100.0%
Budget / size constraints respected	97.9%
Off-topic / injection refused	100.0%
Honest out-of-stock answers	100.0%
Signal extraction accuracy	100.0%
Chat latency p50 / p95	5.5s / 8.8s
Action plan (5 parallel steps incl. Search trends)	5.1s

Eval set: 33 cases, 97.0% passing (English, Hinglish, Tamil, image, budget/size, out-of-stock, off-topic, prompt injection), run against the deployed API. Numbers are reported as measured.

Additional details / future development

Roadmap

Real-time voice conversations with Gemini Live in regional languages

Personalised festive offers from remembered preferences (consented)

Checkout via the Universal Commerce Protocol

Agent-to-agent supplier negotiation for reorders

WhatsApp channel and in-store associate app (store-assist signals)

Shelf photo audits with Gemini vision to catch phantom stock

Responsible AI and path to production

Privacy: signals are aggregated; no personal data in analytics; consent flows aligned with India's DPDP Act 2023

Human in control: no inventory action without manager approval

Pilot: 5 stores × 6 weeks before a festival; success = lower stockout days and recovered lost sales vs control stores

Scale: Pub/Sub signal ingestion, scheduled TimesFM runs for thousands of series, Agent Runtime for long-running agents

Impact and feasibility

For shoppers

Answers in their own language, grounded in real stock

No dead ends: other store, alternative, reserve or notify

Gets the item they actually wanted, in time for the festival

For retailers

Sees demand that sales data hides

Moves stock before the festival peak, not after

Every proposal is explainable: forecast, signals, inventory as evidence

Why it is feasible now

Runs entirely on managed Google Cloud services

No model training: Gemini + TimesFM via BigQuery

Deployed and live today; measured cost and latency

Links

Live prototype (no login)	https://retailnova-ai.web.app
GitHub public repository	https://github.com/tjmanoj/retailnova
Demo video	https://youtu.be/4yQZb8ZcpYQ

Try in 2 minutes: open the live link → “Live Loop” → tap the sample prompt → watch the signal appear → “Generate Diwali action plan” → Approve.

All data is synthetic; Nova Bazaar is a fictional retailer.

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Thank you

